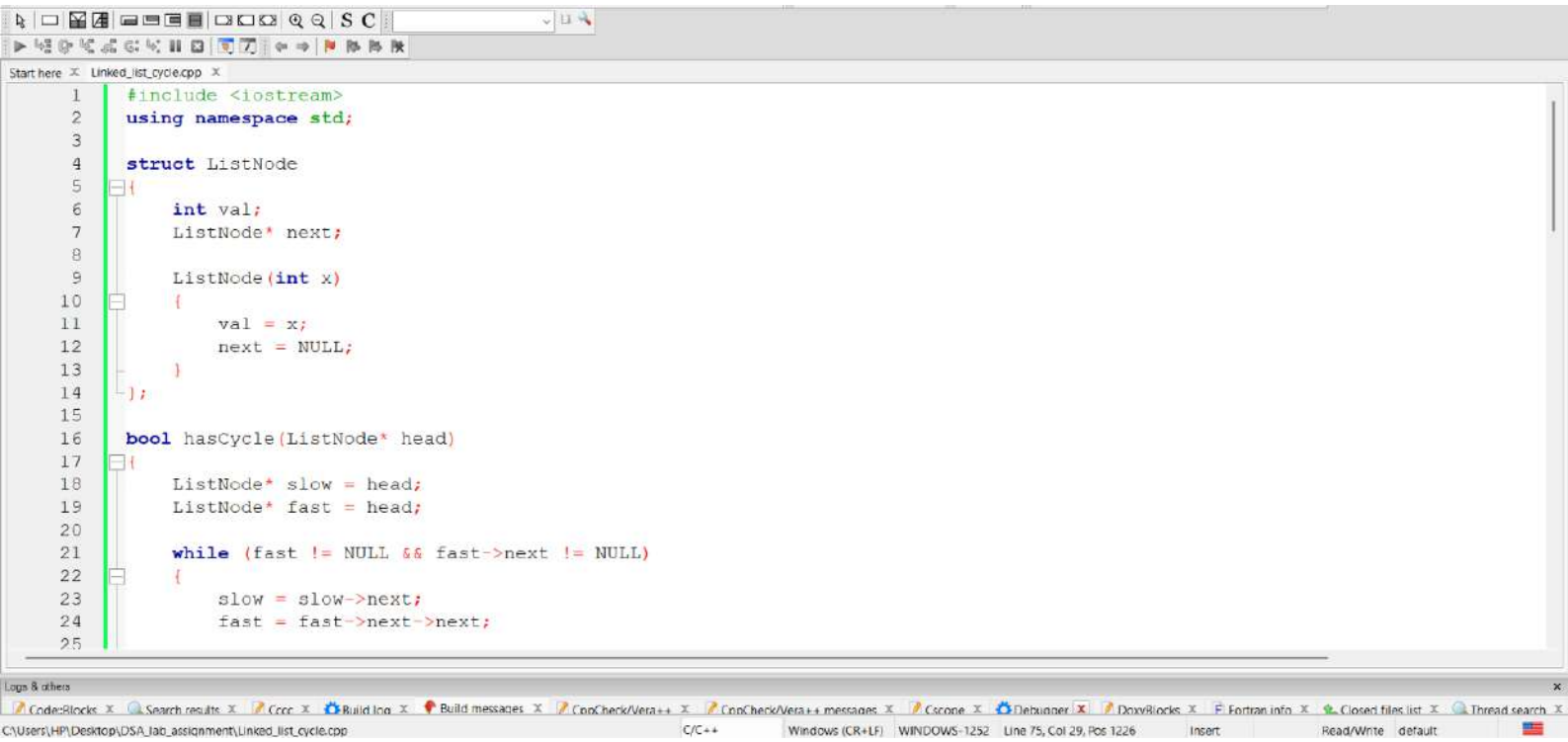


Start here | Linked_list_cycle.cpp

```cpp
#include <iostream>
using namespace std;

struct ListNode
{
    int val;
    ListNode* next;

    ListNode(int x)
    {
        val = x;
        next = NULL;
    }
};

bool hasCycle(ListNode* head)
{
    ListNode* slow = head;
    ListNode* fast = head;

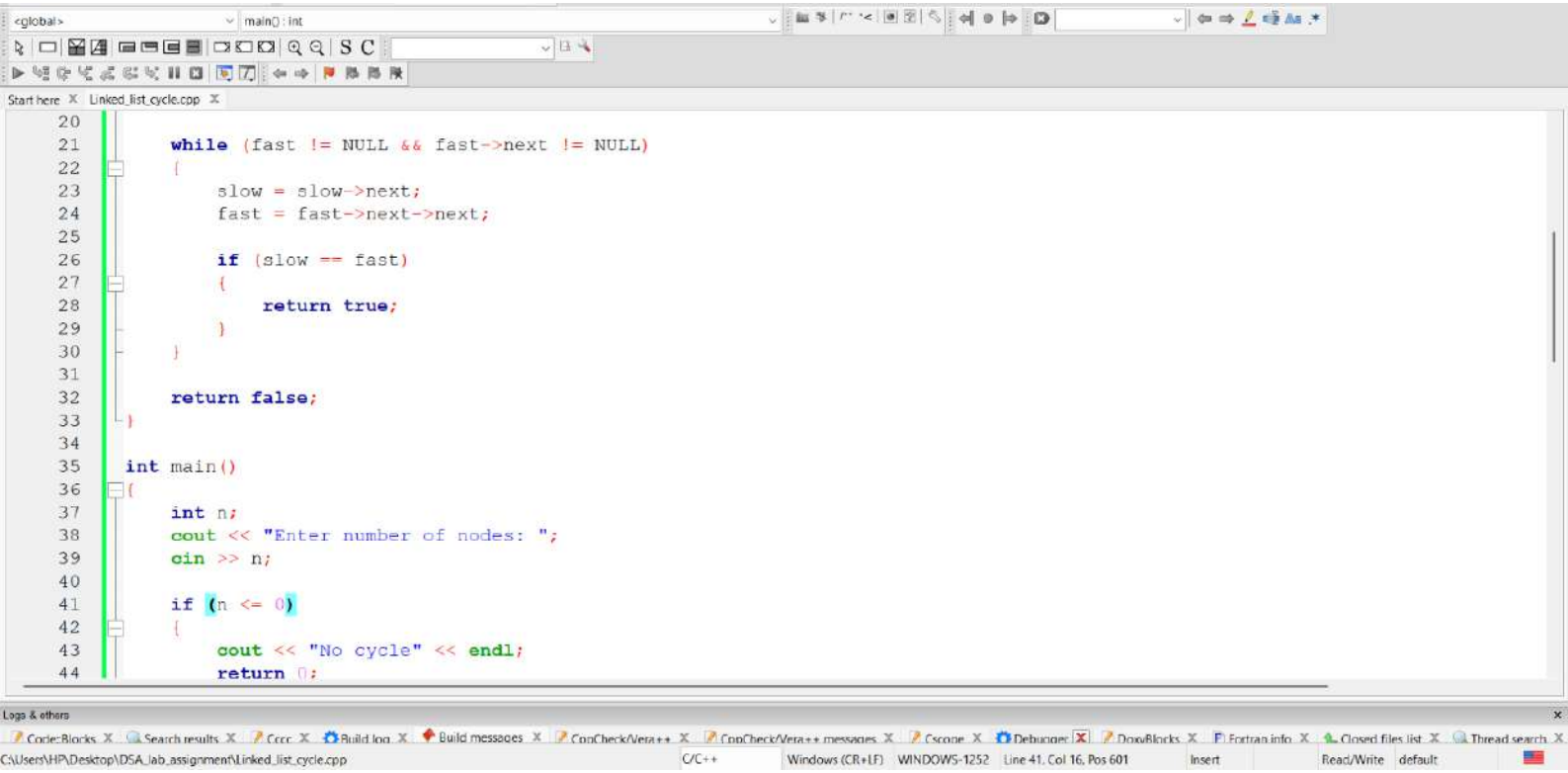
    while (fast != NULL && fast->next != NULL)
    {
        slow = slow->next;
        fast = fast->next->next;

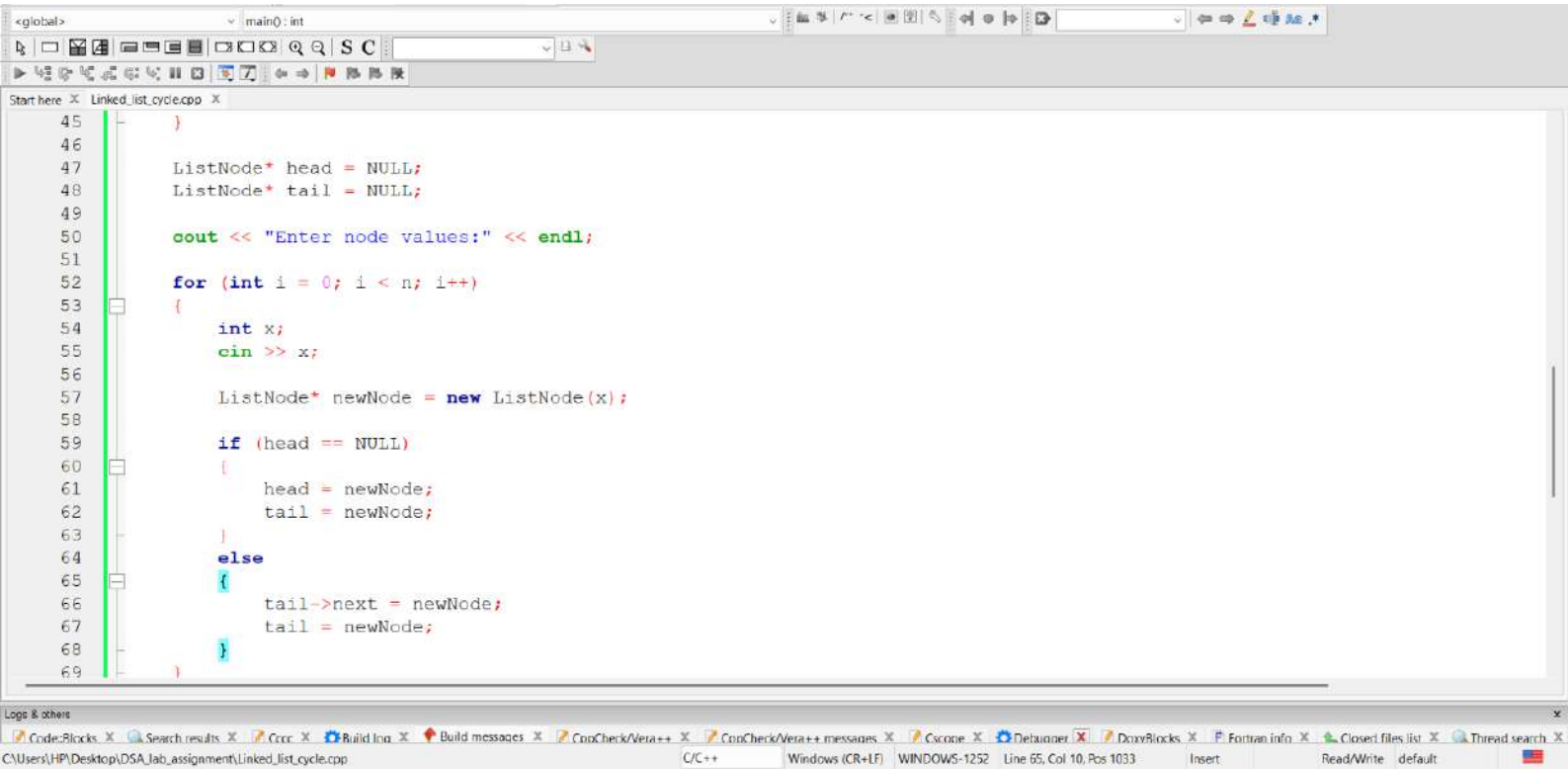
```

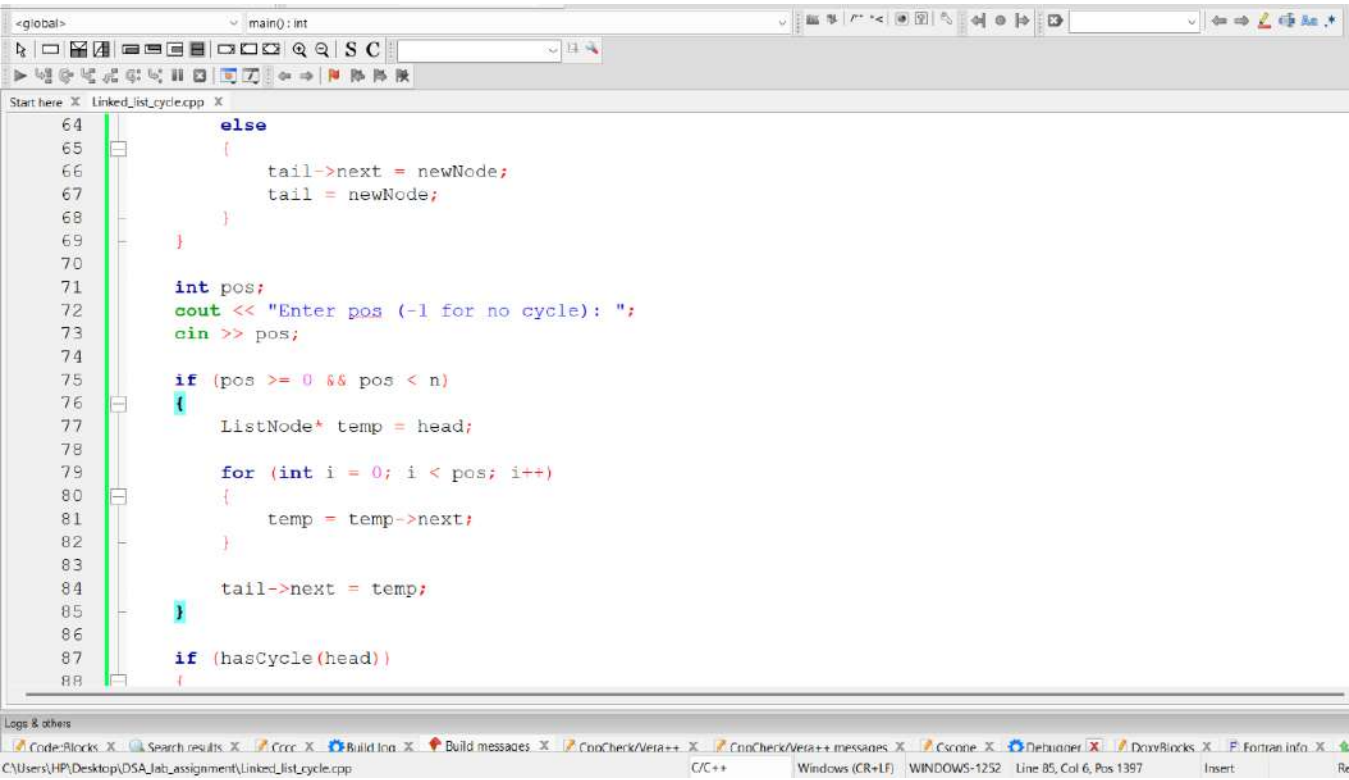
Logs & others

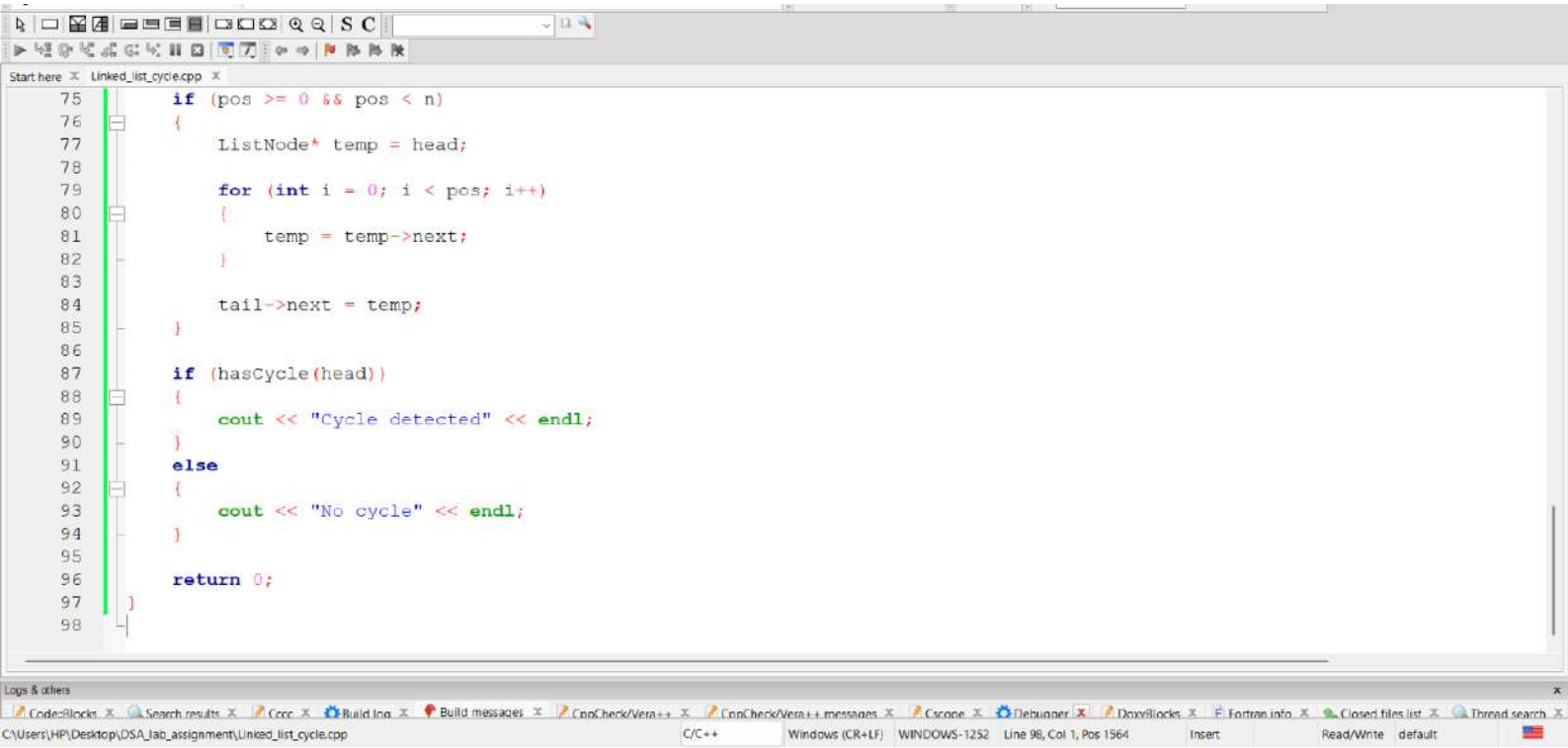
Code::Blocks | Search results | Cccc | Build log | Build messages | CppCheck/Vera++ | CppCheck/Vera++ messages | Cscope | Debugger | DoxyBlocks | Fortran info | Closed files list | Thread search

C:\Users\HP\Desktop\DSA_lab_assignment\Linked_list_cycle.cpp | C/C++ | Windows (CR+LF) | WINDOWS-1252 | Line 75, Col 29, Pos 1226 | Insert | Read/Write | default


Start here
Linked_list_cycle.cpp
20
21 while (fast != NULL && fast->next != NULL)
22 {
23 slow = slow->next;
24 fast = fast->next->next;
25
26 if (slow == fast)
27 {
28 return true;
29 }
30 }
31
32 return false;
33 }
34
35 int main()
36 {
37 int n;
38 cout << "Enter number of nodes: ";
39 cin >> n;
40
41 if (n <= 0)
42 {
43 cout << "No cycle" << endl;
44 return 0;
Logs & others
Code::Blocks
Search results
Cccc
Build log
Build messages
CppCheck/Vera++
CppCheck/Vera++ messages
Cscope
Debugger
DoxyBlocks
Fortran info
Closed files list
Thread search
C:\Users\HP\Desktop\DSA_lab_assignment\Linked_list_cycle.cpp
C/C++
Windows (CR+LF)
WINDOWS-1252
Line 41, Col 16, Pos 601
Insert
Read/Write
default


Start here
Linked_list_cycle.cpp
45 }
46
47 ListNode* head = NULL;
48 ListNode* tail = NULL;
49
50 cout << "Enter node values:" << endl;
51
52 for (int i = 0; i < n; i++)
53 {
54 int x;
55 cin >> x;
56
57 ListNode* newNode = new ListNode(x);
58
59 if (head == NULL)
60 {
61 head = newNode;
62 tail = newNode;
63 }
64 else
65 {
66 tail->next = newNode;
67 tail = newNode;
68 }
69 }
Logs & others
Code::Blocks
Search results
Cccc
Build log
Build messages
CppCheck/Vera++
CppCheck/Vera++ messages
Cscope
Debugger
DoxyBlocks
Fortran info
Closed files list
Thread search
C:\Users\HP\Desktop\DSA_lab_assignment\Linked_list_cycle.cpp
C/C++
Windows (CR+LF)
WINDOWS-1252
Line 65, Col 10, Pos 1033
Insert
Read/Write
default


Start here
Linked_list_cycle.cpp
64 else
65 {
66 tail->next = newNode;
67 tail = newNode;
68 }
69 }
70
71 int pos;
72 cout << "Enter pos (-1 for no cycle): ";
73 cin >> pos;
74
75 if (pos >= 0 && pos < n)
76 {
77 ListNode* temp = head;
78
79 for (int i = 0; i < pos; i++)
80 {
81 temp = temp->next;
82 }
83
84 tail->next = temp;
85 }
86
87 if (hasCycle(head))
88 {
Logs & others
Code::Blocks
Search results
Cccc
Build log
Build messages
CppCheck/Vera++
CppCheck/Vera++ messages
Cscope
Debugger
DoxyBlocks
Fortran info
Closed files list
Thread search
C:\Users\HP\Desktop\DSA_lab_assignment\Linked_list_cycle.cpp
C/C++
Windows (CR+LF)
WINDOWS-1252
Line 85, Col 6, Pos 1397
Insert
Read/Write
default



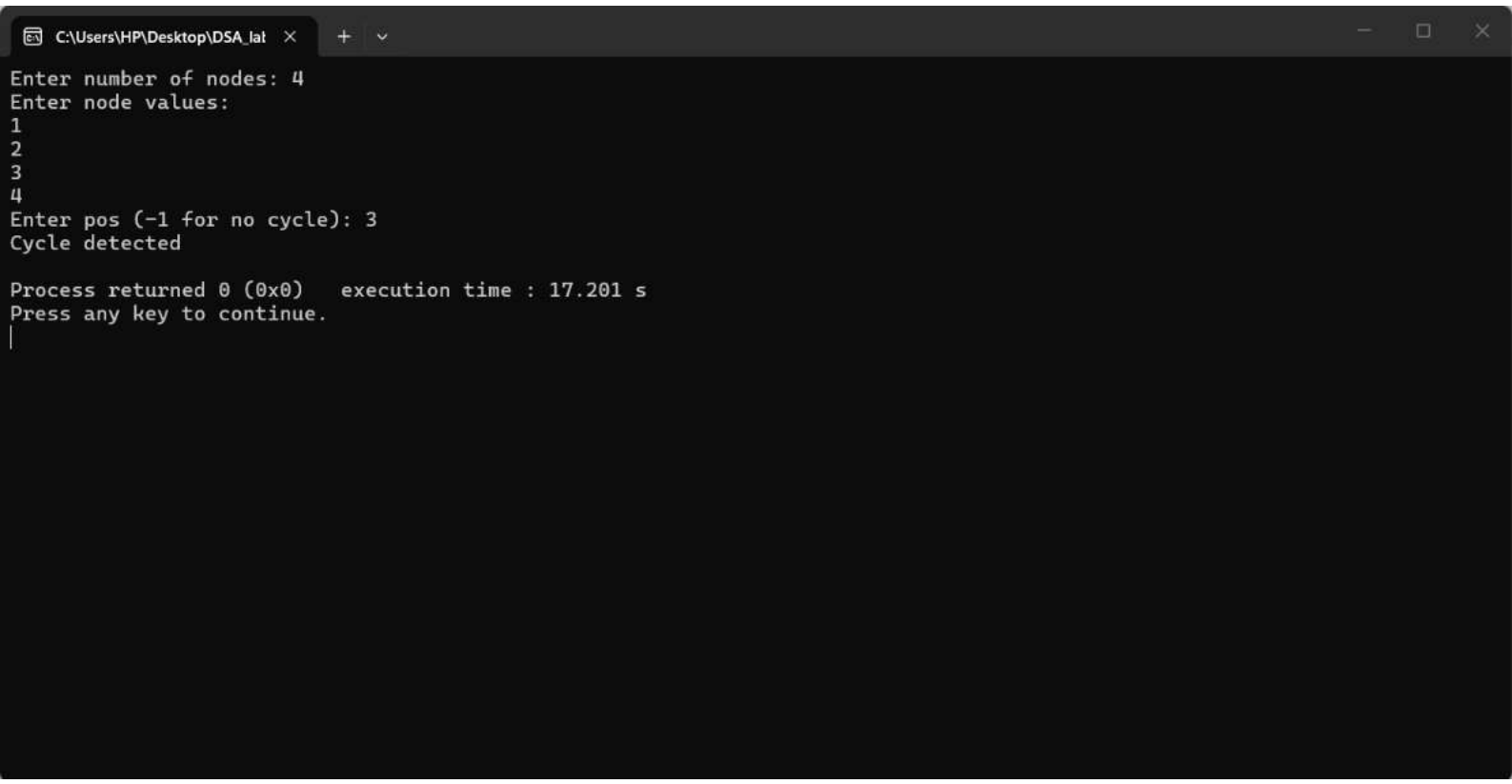
Start here | Linked_list_cycle.cpp

```
75      if (pos >= 0 && pos < n)
76      {
77          ListNode* temp = head;
78
79          for (int i = 0; i < pos; i++)
80          {
81              temp = temp->next;
82          }
83
84          tail->next = temp;
85      }
86
87      if (hasCycle(head))
88      {
89          cout << "Cycle detected" << endl;
90      }
91      else
92      {
93          cout << "No cycle" << endl;
94      }
95
96      return 0;
97  }
98
```

Logs & others

Code::Blocks | Search results | Cccc | Build log | Build messages | CppCheck/Vera++ | CppCheck/Vera++ messages | Cscope | Debugger | DoxyBlocks | Fortran info | Closed files list | Thread search

C:\Users\HP\Desktop\DSA_lab_assignment\Linked_list_cycle.cpp | C/C++ | Windows (CR+LF) | WINDOWS-1252 | Line 98, Col 1, Pos 1564 | Insert | Read/Write | default


C:\Users\HP\Desktop\DSA_lat
Enter number of nodes: 4
Enter node values:
1
2
3
4
Enter pos (-1 for no cycle): 3
Cycle detected
Process returned 0 (0x0)   execution time : 17.201 s
Press any key to continue.



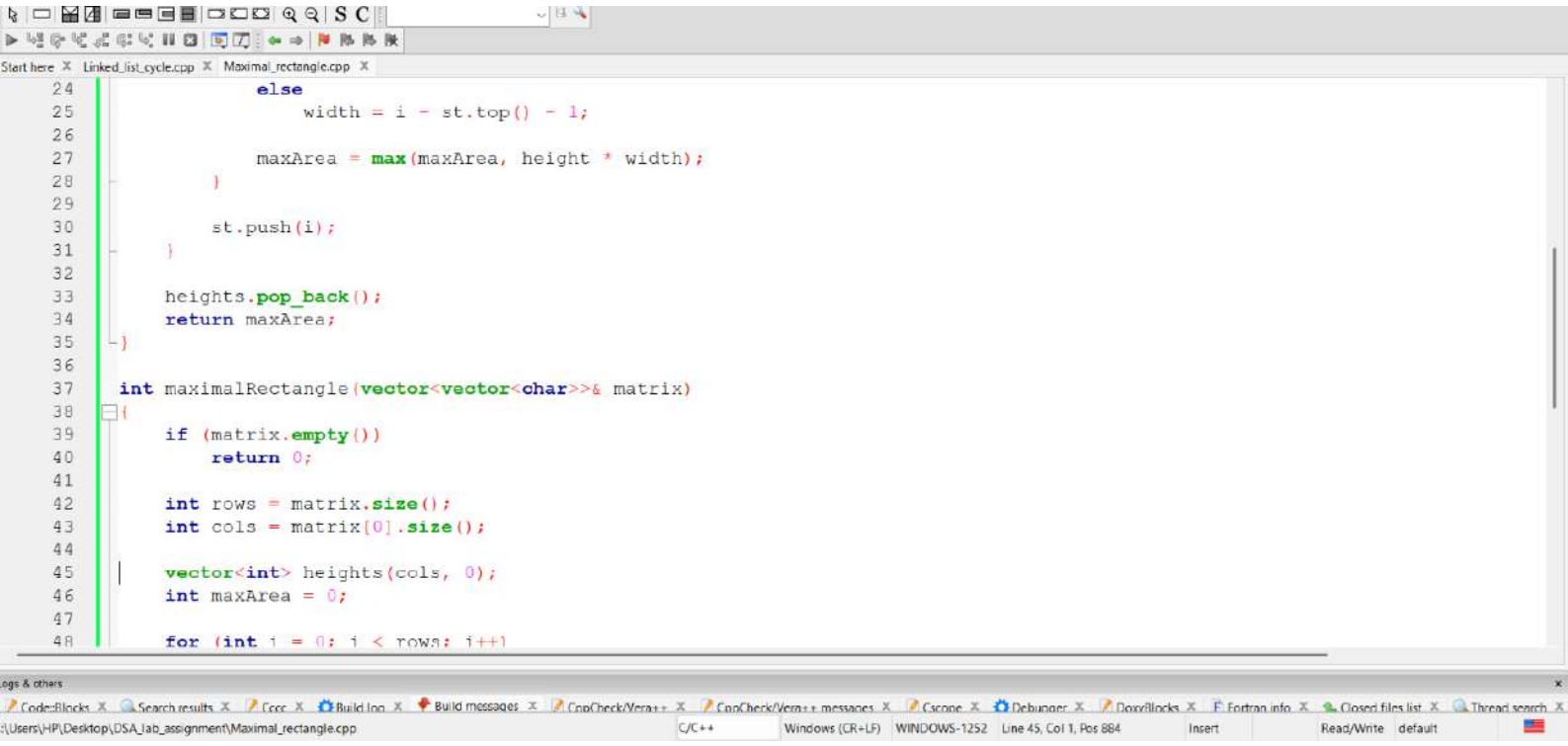
```cpp
#include <iostream>
#include <vector>
#include <stack>
#include <algorithm>
using namespace std;

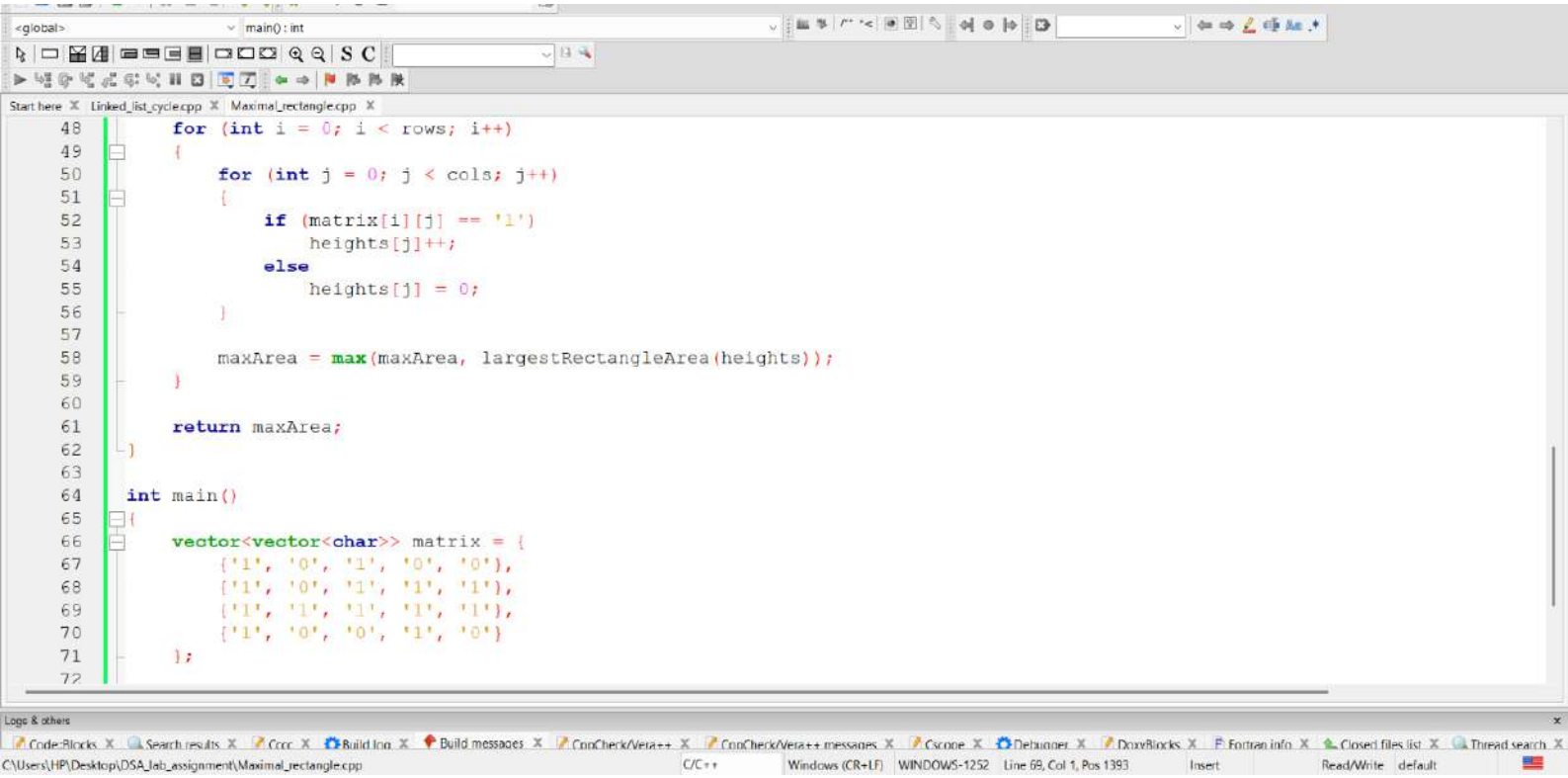
int largestRectangleArea(vector<int>& heights)
{
    stack<int> st;
    int maxArea = 0;
    heights.push_back(0);

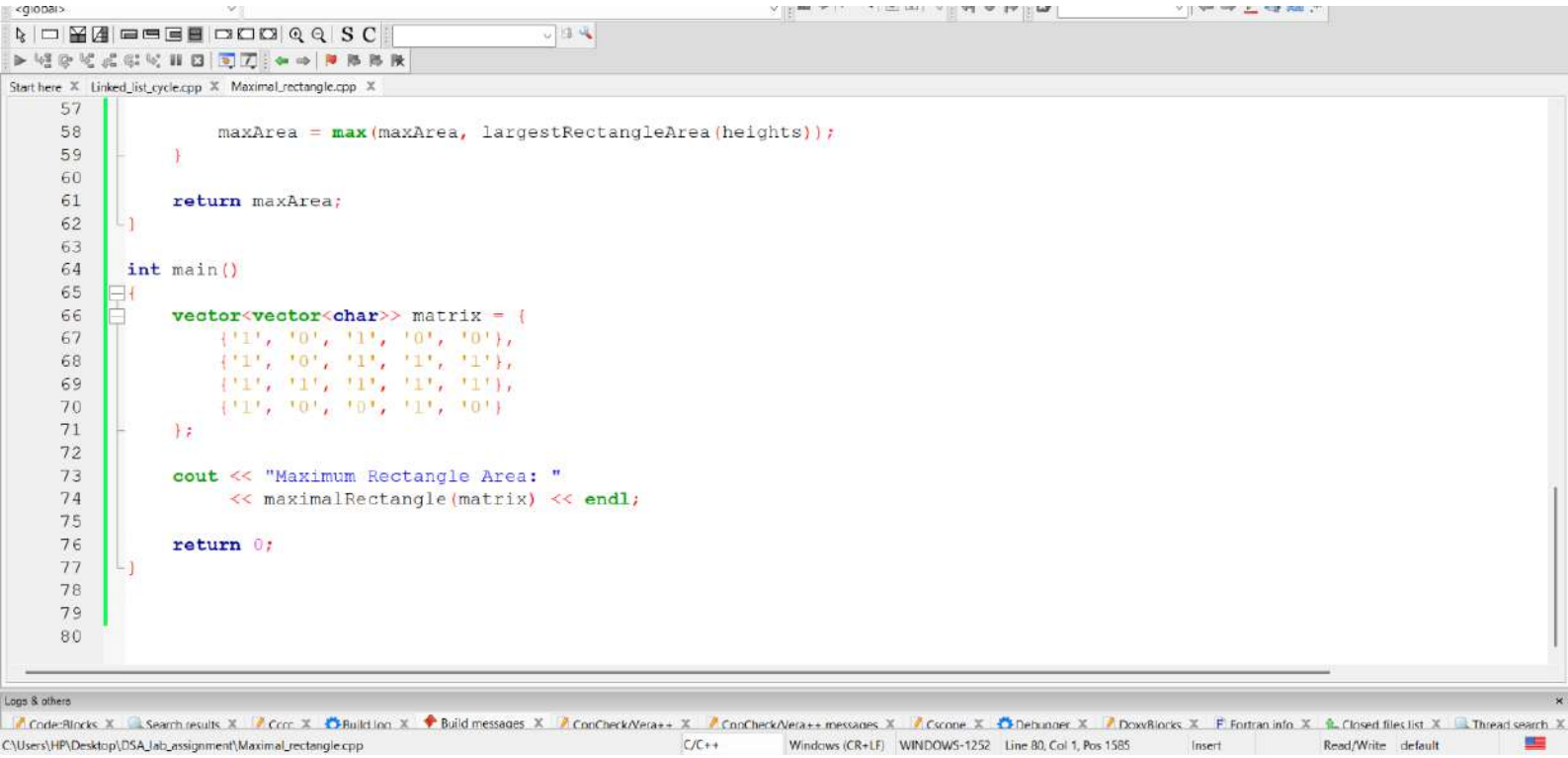
    for (int i = 0; i < heights.size(); i++)
    {
        while (!st.empty() && heights[st.top()] > heights[i])
        {
            int height = heights[st.top()];
            st.pop();

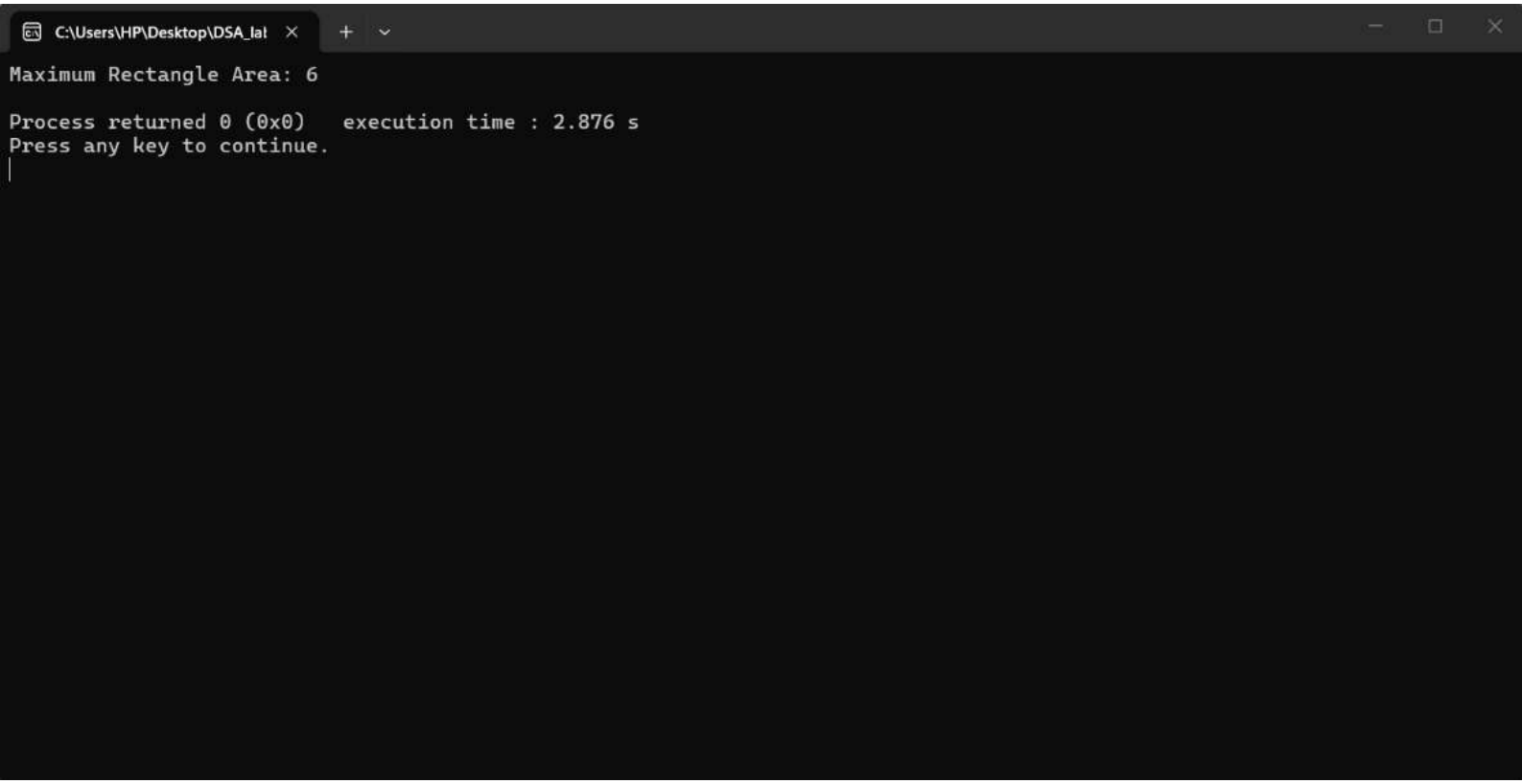
            int width;

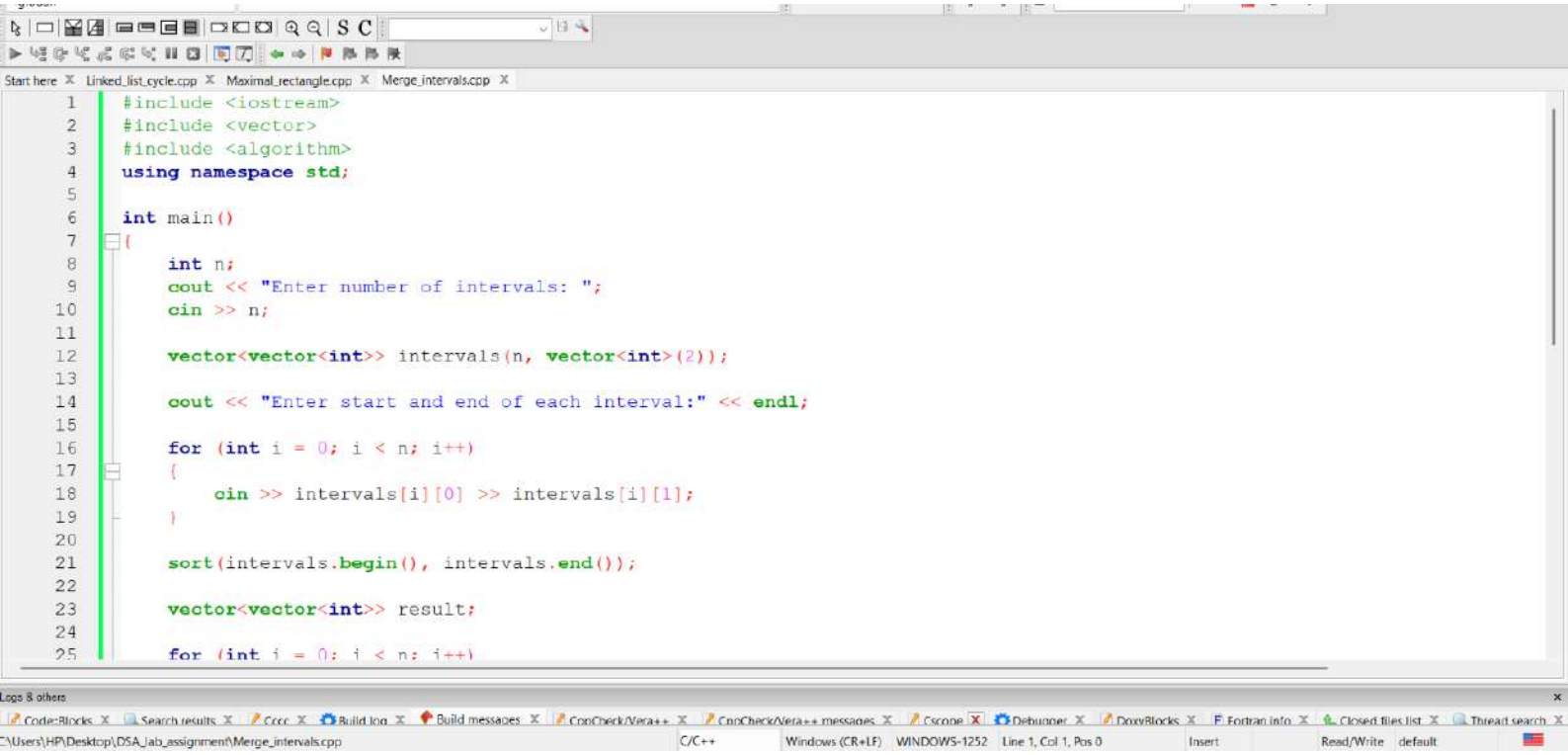
            if (st.empty())
                width = i;
            else
                width = i - st.top() - 1;
```


Start here
Linked_list_cycle.cpp
Maximal_rectangle.cpp
else
width = i - st.top() - 1;
maxArea = max(maxArea, height * width);
}
st.push(i);
}
heights.pop_back();
return maxArea;
}
int maximalRectangle(vector<vector<char>>& matrix)
{
if (matrix.empty())
return 0;
int rows = matrix.size();
int cols = matrix[0].size();
vector<int> heights(cols, 0);
int maxArea = 0;
for (int i = 0; i < rows; i++)
Logs & others
Code::Blocks
Search results
Cccc
Build log
Build messages
CppCheck/Vera++
CppCheck/Vera++ messages
Cscope
Debugger
DoxyBlocks
Fortran info
Closed files list
Thread search
:\Users\HP\Desktop\DSA_lab_assignment\Maximal_rectangle.cpp
C/C++
Windows (CR+LF)
WINDOWS-1252
Line 45, Col 1, Pos 884
Insert
Read/Write
default


Start here
Linked_list_cycle.cpp
Maximal_rectangle.cpp
48 for (int i = 0; i < rows; i++)
49 {
50 for (int j = 0; j < cols; j++)
51 {
52 if (matrix[i][j] == '1')
53 heights[j]++;
54 else
55 heights[j] = 0;
56 }
57
58 maxArea = max(maxArea, largestRectangleArea(heights));
59 }
60
61 return maxArea;
62 }
63
64 int main()
65 {
66 vector<vector<char>> matrix = {
67 {'1', '0', '1', '0', '0'},
68 {'1', '0', '1', '1', '1'},
69 {'1', '1', '1', '1', '1'},
70 {'1', '0', '0', '1', '0'}
71 };
72
Logs & others
Code::Blocks
Search results
Cccc
Build log
Build messages
CppCheck/Vera++
CppCheck/Vera++ messages
Cscope
Debugger
DoxyBlocks
Fortran info
Closed files list
Thread search
C:\Users\HP\Desktop\DSA_lab_assignment\Maximal_rectangle.cpp
C/C++
Windows (CR+LF)
WINDOWS-1252
Line 69, Col 1, Pos 1393
Insert
Read/Write
default


Start here
Linked_list_cycle.cpp
Maximal_rectangle.cpp
maxArea = max(maxArea, largestRectangleArea(heights));
}
return maxArea;
}
int main()
{
vector<vector<char>> matrix = {
{'1', '0', '1', '0', '0'},
{'1', '0', '1', '1', '1'},
{'1', '1', '1', '1', '1'},
{'1', '0', '0', '1', '0'}
};
cout << "Maximum Rectangle Area: "
<< maximalRectangle(matrix) << endl;
return 0;
}
Logs & others
Code::Blocks
Search results
Cccc
Build log
Build messages
CppCheck/Vera++
CppCheck/Vera++ messages
Cscope
Debugger
DoxyBlocks
Fortran info
Closed files list
Thread search
C:\Users\HP\Desktop\DSA_lab_assignment\Maximal_rectangle.cpp
C/C++
Windows (CR+LF)
WINDOWS-1252
Line 80, Col 1, Pos 1585
Insert
Read/Write
default


C:\Users\HP\Desktop\DSA_lal
Maximum Rectangle Area: 6
Process returned 0 (0x0)   execution time : 2.876 s
Press any key to continue.



Start here | Linked_list_cycle.cpp | Maximal_rectangle.cpp | Merge_intervals.cpp

```cpp
#include <iostream>
#include <vector>
#include <algorithm>
using namespace std;

int main()
{
    int n;
    cout << "Enter number of intervals: ";
    cin >> n;

    vector<vector<int>> intervals(n, vector<int>(2));

    cout << "Enter start and end of each interval:" << endl;

    for (int i = 0; i < n; i++)
    {
        cin >> intervals[i][0] >> intervals[i][1];
    }

    sort(intervals.begin(), intervals.end());

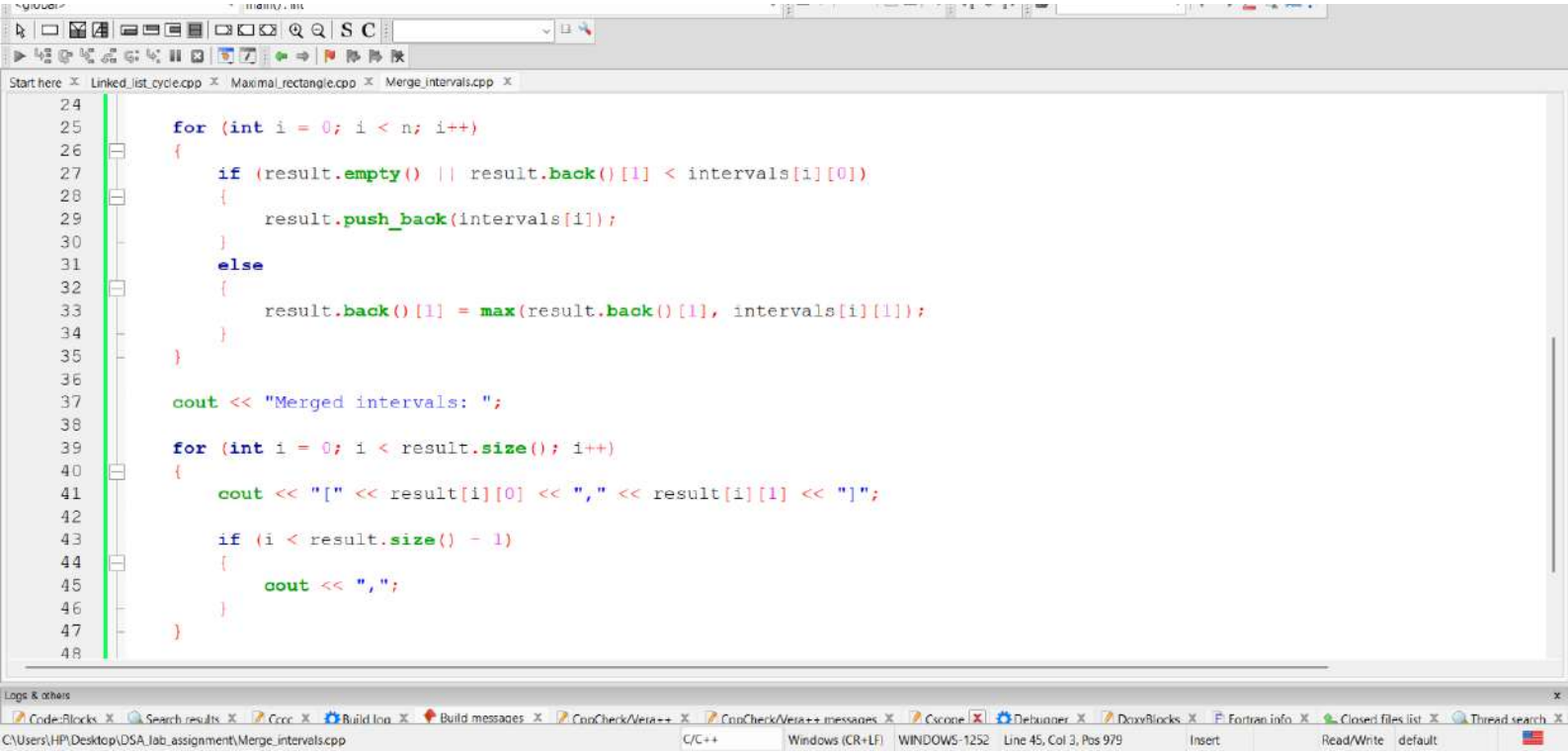
    vector<vector<int>> result;

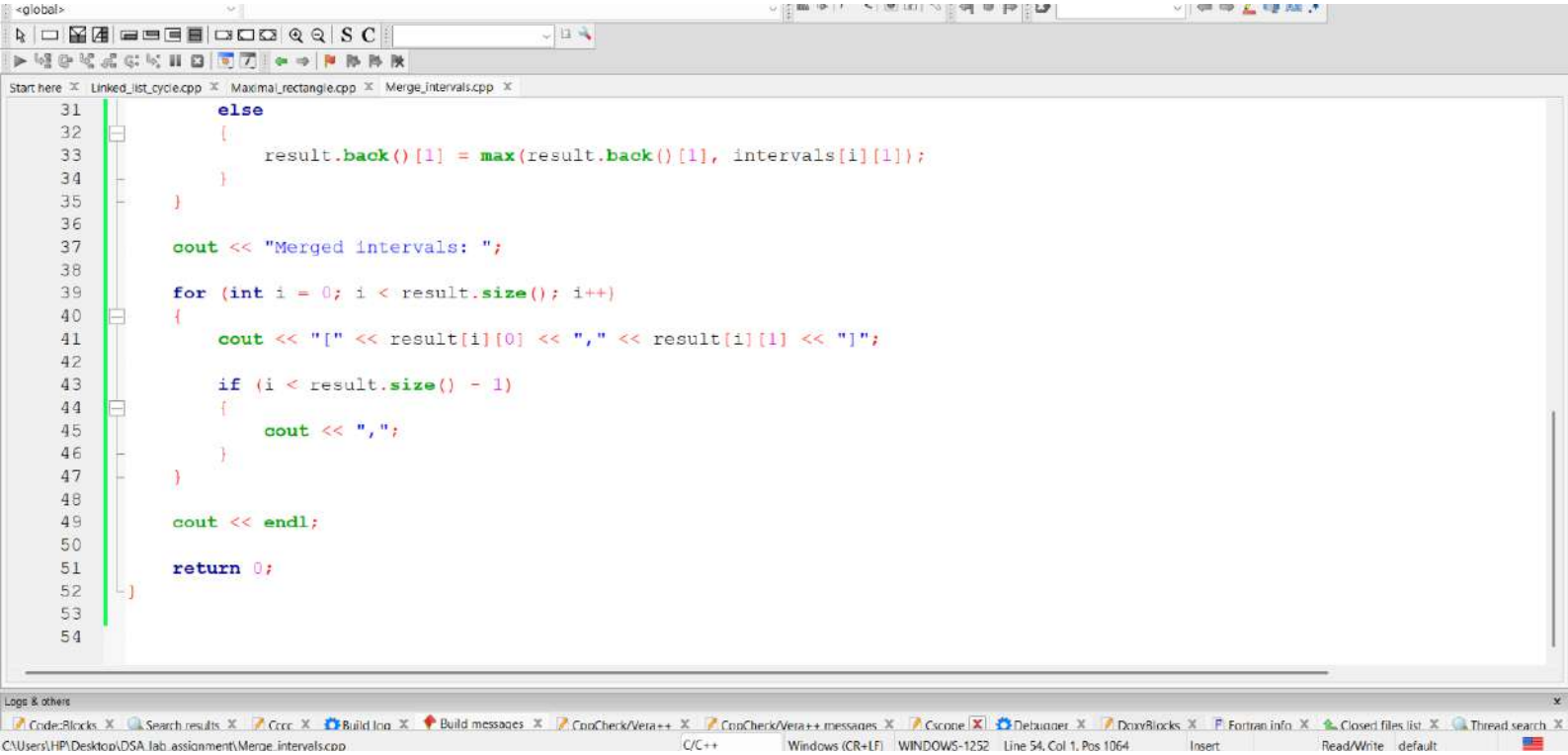
    for (int i = 0; i < n; i++)
```

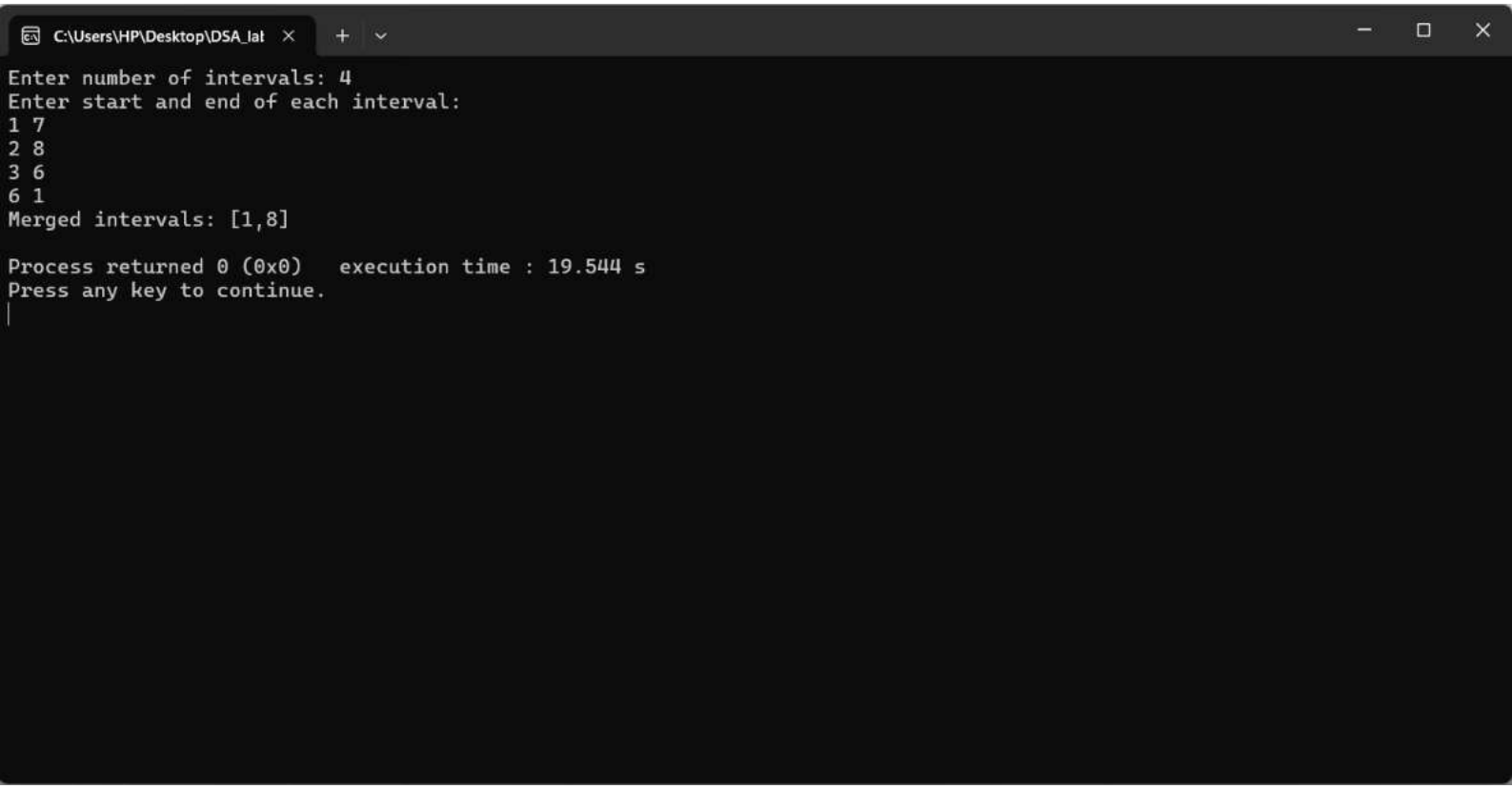
Logs & others

Code::Blocks | Search results | Cccc | Build log | Build messages | CppCheck/Vera++ | CppCheck/Vera++ messages | Cscope | Debugger | DoxyBlocks | Fortran info | Closed files list | Thread search

C:\Users\HP\Desktop\DSA_lab_assignment\Merge_intervals.cpp | C/C++ | Windows (CR+LF) | WINDOWS-1252 | Line 1, Col 1, Pos 0 | Insert | Read/Write | default


Start here
Linked_list_cycle.cpp
Maximal_rectangle.cpp
Merge_intervals.cpp
for (int i = 0; i < n; i++)
{
if (result.empty() || result.back()[1] < intervals[i][0])
{
result.push_back(intervals[i]);
}
else
{
result.back()[1] = max(result.back()[1], intervals[i][1]);
}
}
cout << "Merged intervals: ";
for (int i = 0; i < result.size(); i++)
{
cout << "[" << result[i][0] << "," << result[i][1] << "]";
if (i < result.size() - 1)
{
cout << ",";
}
}
Logs & others
Code::Blocks
Search results
Cccc
Build log
Build messages
CppCheck/Vera++
CppCheck/Vera++ messages
Cscope
Debugger
DoxyBlocks
Fortran info
Closed files list
Thread search
C:\Users\HP\Desktop\DSA_lab_assignment\Merge_intervals.cpp
C/C++
Windows (CR+LF)
WINDOWS-1252
Line 45, Col 3, Pos 979
Insert
Read/Write
default


Start here
Linked_list_cycle.cpp
Maximal_rectangle.cpp
Merge_intervals.cpp
else
{
result.back()[1] = max(result.back()[1], intervals[i][1]);
}
}
cout << "Merged intervals: ";
for (int i = 0; i < result.size(); i++)
{
cout << "[" << result[i][0] << "," << result[i][1] << "]";
if (i < result.size() - 1)
{
cout << ",";
}
}
cout << endl;
return 0;
}
Logs & others
Code::Blocks
Search results
Cccc
Build log
Build messages
CppCheck/Vera++
CppCheck/Vera++ messages
Cscope
Debugger
DoxyBlocks
Fortran info
Closed files list
Thread search
C:\Users\HP\Desktop\DSA lab assignment\Merge_intervals.cpp
C/C++
Windows (CR+LF)
WINDOWS-1252
Line 54, Col 1, Pos 1064
Insert
Read/Write
default


C:\Users\HP\Desktop\DSA_lat
Enter number of intervals: 4
Enter start and end of each interval:
1 7
2 8
3 6
6 1
Merged intervals: [1,8]
Process returned 0 (0x0)   execution time : 19.544 s
Press any key to continue.

Start here | Linked_list_cycle.cpp | Maximal_rectangle.cpp | *Merge_intervals.cpp | Number_of_islands.cpp

```cpp
#include <iostream>
#include <vector>
using namespace std;

void dfs(vector<vector<char>>& grid, int i, int j, int m, int n)
{
    if (i < 0 || i >= m || j < 0 || j >= n || grid[i][j] == '0')
    {
        return;
    }

    grid[i][j] = '0';

    dfs(grid, i + 1, j, m, n);
    dfs(grid, i - 1, j, m, n);
    dfs(grid, i, j + 1, m, n);
    dfs(grid, i, j - 1, m, n);
}

int main()
{
    int m, n;

    cout << "Enter rows and columns: ";
    cin >> m >> n;

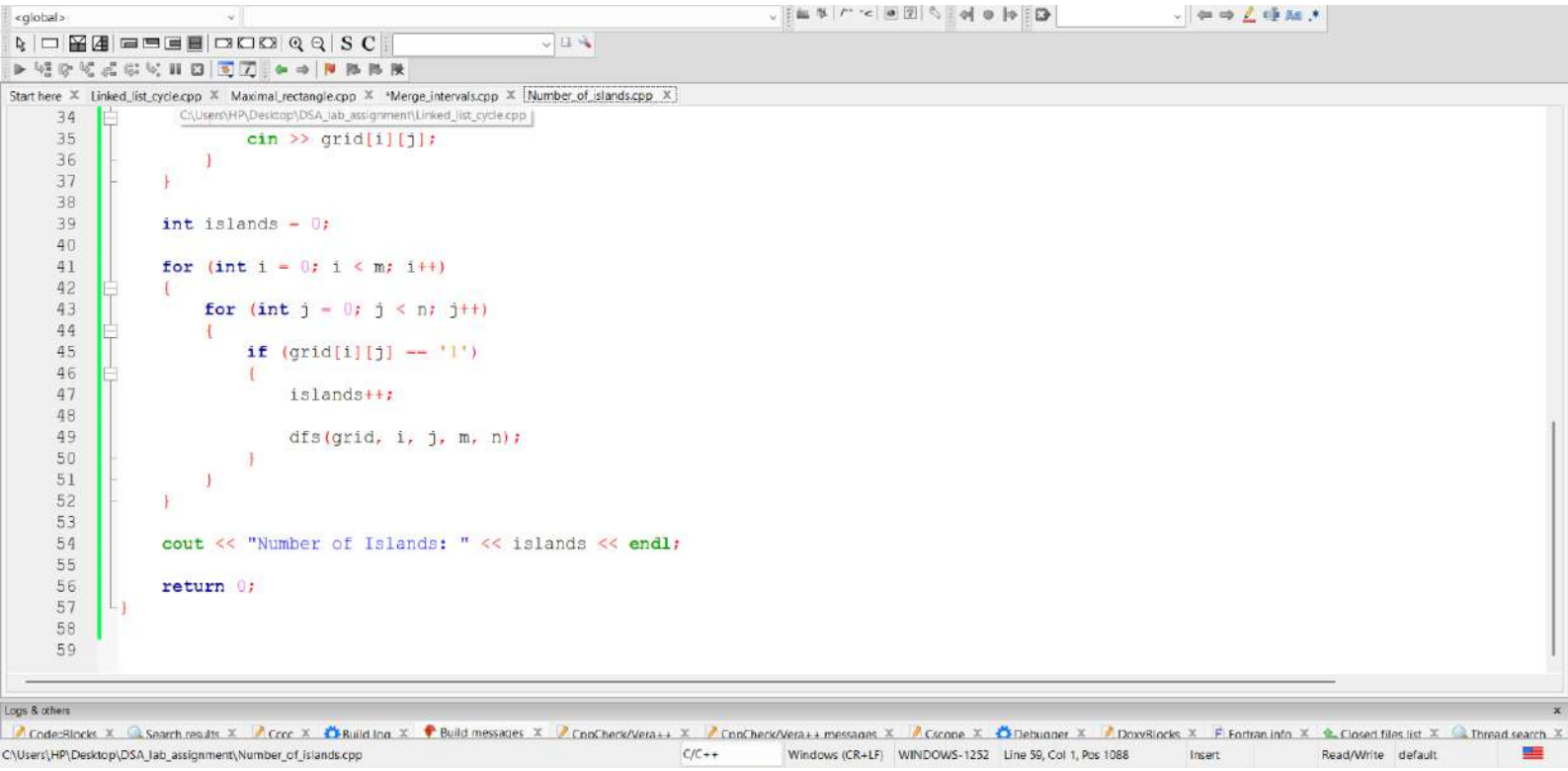
    vector<vector<char>> grid(m, vector<char>(n));
```

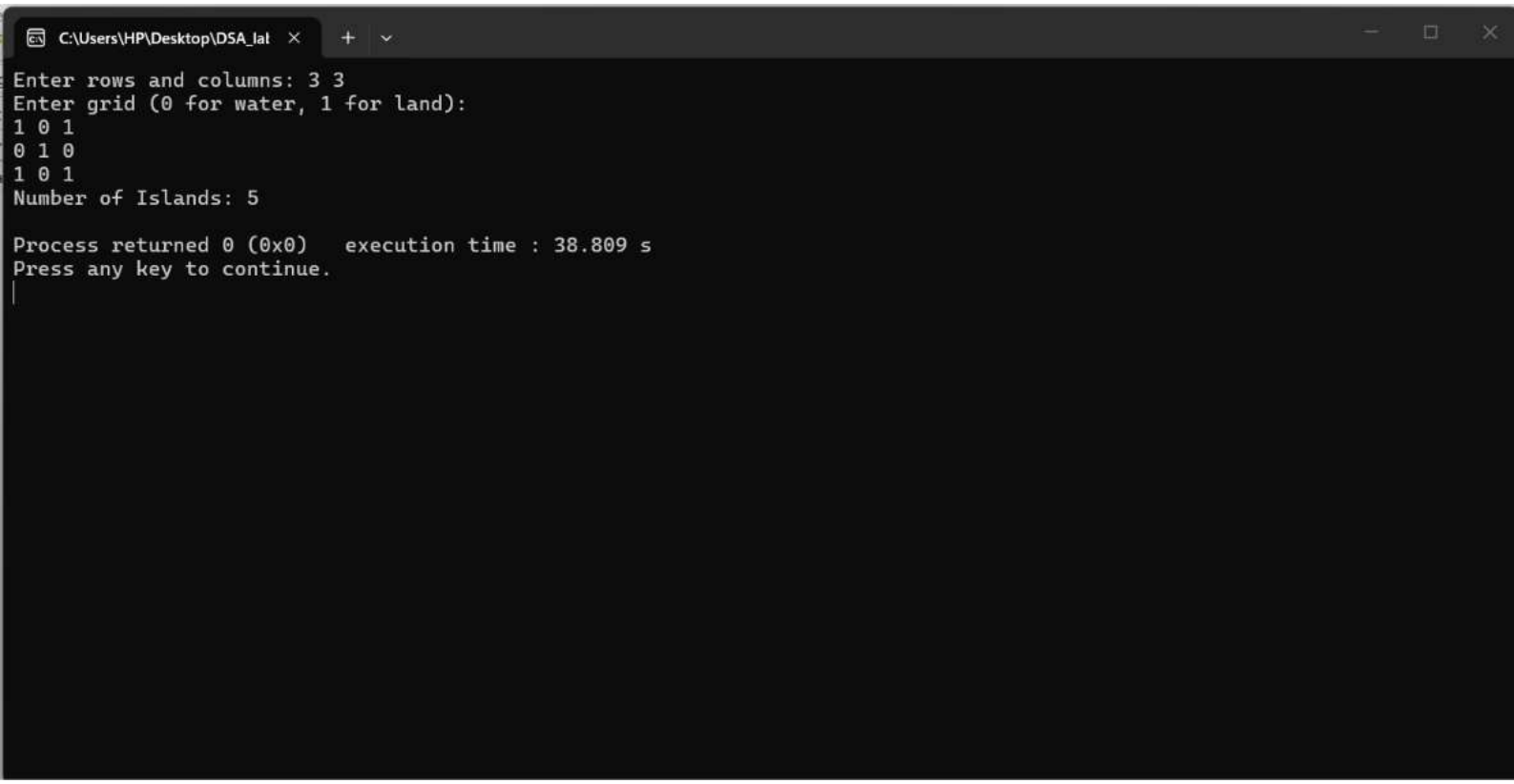
Logs & others

Code::Blocks | Search results | Cccc | Build log | Build messages | CppCheck/Vera++ | CppCheck/Vera++ messages | Cscope | Debugger | DoxyBlocks | Fortran info | Closed files list | Thread search

C:\Users\HP\Desktop\DSA_lab_assignment\Number_of_islands.cpp | C/C++ | Windows (CR+LF) | WINDOWS-1252 | Line 58, Col 1, Pos 1086 | Insert | Read/Write | default


art here
Linked_list_cycle.cpp
Maximal_rectangle.cpp
*Merge_intervals.cpp
Number_of_islands.cpp
19
20 int main()
21 {
22 int m, n;
23
24 cout << "Enter rows and columns: ";
25 cin >> m >> n;
26
27 vector<vector<char>> grid(m, vector<char>(n));
28
29 cout << "Enter grid (0 for water, 1 for land):" << endl;
30
31 for (int i = 0; i < m; i++)
32 {
33 for (int j = 0; j < n; j++)
34 {
35 cin >> grid[i][j];
36 }
37 }
38
39 int islands = 0;
40
41 for (int i = 0; i < m; i++)
42 {
43 for (int j = 0; j < n; j++)
44 {
45 if (grid[i][j] == '1')
& others
Code::Blocks
Search results
Cccc
Build log
Build messages
CppCheck/Vera++
CppCheck/Vera++ messages
Cscope
Debugger
DoxyBlocks
Fortran info
Closed files list
Thread search
sers\HP\Desktop\DSA_lab_assignment\Number_of_islands.cpp
C/C++
Windows (CR+LF)
WINDOWS-1252
Line 42, Col 2, Pos 801
Insert
Read/Write
default


Start here
Linked_list_cycle.cpp
Maximal_rectangle.cpp
*Merge_intervals.cpp
Number_of_islands.cpp
C:\Users\HP\Desktop\DSA_lab_assignment\Linked_list_cycle.cpp
cin >> grid[i][j];
int islands = 0;
for (int i = 0; i < m; i++)
for (int j = 0; j < n; j++)
if (grid[i][j] == '1')
islands++;
dfs(grid, i, j, m, n);
cout << "Number of Islands: " << islands << endl;
return 0;
Logs & others
C:\Users\HP\Desktop\DSA_lab_assignment\Number_of_islands.cpp
C/C++
Windows (CR+LF)
WINDOWS-1252
Line 59, Col 1, Pos 1088
Insert
Read/Write
default


C:\Users\HP\Desktop\DSA_lal
Enter rows and columns: 3 3
Enter grid (0 for water, 1 for land):
1 0 1
0 1 0
1 0 1
Number of Islands: 5
Process returned 0 (0x0)   execution time : 38.809 s
Press any key to continue.